INCH-POUND
MIL-DTL-0032175 (AR)
26 April 2004

DETAIL SPECIFICATION

FUZE, PIBD - M549A1, Less Spitback Assembly

This specification is approved for interim use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the metal parts, assemblies, loading and assembling for one type of fuze designated as FUZE, PIBD: M549A1, Less Spitback Assembly, hereafter referred to as the fuze, or fuze assembly.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-5541 - Chemical Conversion Coating on Aluminum and Aluminum Alloys
MIL-A-8625 - Anodic Coatings for Aluminum and Aluminum Alloys
MIL-D-14978 - Detonator, Stab, M55 Loading, Assembling and Packaging
MIL-A-70625 - Automated acceptance inspection equipment design, Testing and Approval of

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, New Jersey 07806-5000, or e-mailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-1168 - Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

U.S. (ARDEC) DRAWINGS (see 6.5)

PRODUCT AND PACKAGING DRAWINGS

8864602 - Hammerweight Pin
8876226 - Bracket Spring
8886368 - Pinion assembly
8886369 - Pinion
8886370 - Starwheel
9129876 - Rotor shaft
9219872 - Sector gear
9287862 - Spitback Assembly
9287882 - Setback Pin
9287890 - Bottom plate
12944723 - Centerplate
12944724 - Bracket
12944725 - Escapement Assembly
12944726 - Escapement Housing
12944727 - Verge Assembly
12944728 - Verge Arbor
12944729 - Rotor Assembly
12944730 - Rotor and Gear Sector Assembly
12944731 - Rotor
12944732 - Top Plate
12944733 - Spinlock
12944734 - Centerplate Assembly
12944735 - Ogive
12944737 - Spinlock Spring
12944738 - Fuze, PIBD, M549A1, Less Spitback
12944740 - Ring
12944741 - Setback Spring
12944744 - Verge
12944747 - Fuze, PIBD, M549A1:
12944746 - Firing Pin
12944748 - Bushing
12944749 - Firing Pin Spring
12993604 - Cartridge, 40mm, Test Vehicle with M549A1 Fuze

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12993605 - Test Vehicle, w/Ctrg Case M169 for M549A1 Fuze

INSPECTION EQUIPMENT DRAWINGS

9202253 - Chamber Gage
9202254 - Gage Plug Wear Limit Check
9202255 - Gage Plug Acceptance Check

(Copies of these drawings may be requested online at drawing-request@pica.army.mil, or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS INTERNATIONAL (ASTM)

ASTM B117 - Standard Method of Salt Spray (Fog) Testing
ASTM C208 - Insulation Board (Cellulosic fiber), Structural and Decorative
ASTM E1255 - Standard Practice for Radioscopy

(Copies of this document are available online at www.astm.org or from the ASTM International, 100 Barr Harbor Drive, Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the solicitation or contract order (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials, parts, and assemblies shall comply with requirements specified on the applicable drawings and referenced specifications.

3.3 Transportation vibration. Prior to transportation vibration test, the fuze assembly shall be x-rayed and shall not be armed or partially armed (as evidenced by the detonator moving any degree from the safe position) or have any safety feature missing or not in its proper position (i.e. both spin locks/spring combination missing, pinion/verge assembly missing). After performance of the transportation vibration test the fuze assembly shall be x-rayed again and shall not be armed or partially armed, (as evidenced by the detonator moving any degree from the safe position) have any safety feature not in its proper position, damaged parts, foreign material and shall be safe for handling and disposal

3.4 Jolt. The fuze assembly shall be x-rayed and shall not be armed or partially armed (as evidenced by the detonator moving any degree from safe position) or have any safety feature missing or not in its proper position (i.e. both spin locks/spring combination missing, pinion/verge assembly missing). After being subjected to the jolt test, the fuze assembly shall be

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x-rayed again and shall not be armed or partially armed, (as evidenced by the detonator moving any degree from the safe position) have any safety feature not in its proper position, damaged parts, foreign material and shall be safe for handling and disposal

3.5 Jumble. The fuze assembly shall be x-rayed and shall not be armed or partially armed (as evidenced by the detonator moving any degree from the safe position) or have any safety feature missing or not in its proper position (i.e. both spin locks/spring combination missing, pinion/verge assembly missing). After being subjected to the jumble test, the fuze assembly shall be x-rayed again and shall not be armed or partially armed, (as evidenced by the detonator moving any degrees from the safe position) have any safety feature not in its proper position, damaged parts, foreign material and shall be safe for handling and disposal

3.6 Fuze all arming/graze sensitivity. The fuze assembly shall function high order as evidenced by the emittance of a visual (smoke or flash) or audible signature when it strikes the specified impact medium.

3.7 Fuze non-Arming. The fuze assembly shall not function as evidenced by the lack of a visual (smoke and flash) or audible signature when it strikes the specified target.

3.8 Premature burst. The fuze assembly shall not function in the gun tube or in uninterrupted flight.

3.9 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough and workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coating shall be continuous; however, a few light scratches not exposing base material may be permitted. Rack marks on the ogive are permissible. All required marking and stamping shall be neat and sharply defined.

3.10 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

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4. VERIFICATION

TABLE I. Requirement/Inspection Cross Reference Matrix

Method of inspection
N/A – Not Applicable (N/A)
1 – Analysis
2 – Demonstration
3 – Examination
4 – Test

Classes of inspection
A – First article
B – Conformance

SECTION 3 REQUIREMENT	INSPECTION METHODS					INSPECTION CLASS		SECTION 4 INSPECTION METHOD
	N/A	1	2	3	4	A	B	
3.1			X	X	X	X		4.3
3.2			X	X	X	X	X	4.4
3.3			X	X	X	X	X	4.5.21/4.5.20
3.4			X	X	X	X	X	4.5.22/4.5.20
3.5			X	X	X	X	X	4.5.23/4.5.20
3.6			X		X	X	X	4.5.24/4.5.25
3.7			X		X	X	X	4.5.26
3.8				X	X	X	X	4.5.28
3.9				X		X	X	4.5.33
3.10				X		X	X	4.5.6

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3)
b. Conformance Inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in section 4.5.

4.3 First article. When specified in the solicitation or contract, a sample of the Fuze, PIBD: M549A1, less spitback assembly shall be subjected to first article inspection in accordance with Table III and 4.3.1.

4.3.1 First article quantity. The contractor shall submit a first article sample for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated in Table II.

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TABLE II. First Article Quantity

PART DESCRIPTION	DRAWINGS	QUANTITY
Escapement Housing	12944726	25
Verge Arbor	12944728	25
Pinion	8886369	25
Ogive	12944735	25
Firing Pin	12944746	25
Setback Pin	9287882	125
Setback Spring	12944741	125
Starwheel	8886370	25
Bushing	12944748	25
Pinion assembly	8886368	125
Verge Assembly	12944727	125
Top Plate	12944732	25
Rotor and Gear Sector Assembly	12944730	125
Rotor Assembly	12944729	125
Rotor	12944731	25
Hammerweight Pin	8864602	25
Escapement Assembly Less Setback Pin and Setback Spring	12944725 (Less 9287882 and 12944741)	125
Escapement Assembly	12944725	125
Spinlock	12944733	25
Bottom plate	9287890	25
Rotor shaft	9129876	25
Firing Pin Spring	12944749	25
Verge	12944744	25
Spinlock Spring	12944737	25
Centerplate Assembly	12944734	125
Ring	12944740	25
Centerplate	12944723	25
Bracket	12944724	25
Bracket Spring	8876226	25
Sector gear	9219872	25
Fuze, PIBD, M549A1, Less Spitback	12944738	508

4.3.2 Inspection to be performed. The first article inspection shall be 100% examination of all drawing characteristics (listed and unlisted) including workmanship requirements for the parts and assemblies listed in 4.3.1 and the tests specified in Table III herein.

4.3.3 First article rejection. If any item of the sample fails to comply with the first article requirements, the sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE III. First Article Inspection

FUZE, PIBD, M549A1 Less Spitback and Components 1 of 2			
EXAMINATION OR TEST	NO. OF SAMPLE UNITS	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Setback Spring (Dwg. 12944741)			
Push off test	125	3.2	4.5.4
Verge Assembly (Dwg. 12944727)			
Torque test	50	3.2	4.5.8
Disassembly test	50	3.2	4.5.9
Pinion Assembly (Dwg. 8886368)			
Push off test	50	3.2	4.5.10
Torque test	50	3.2	4.5.11
Rotor Assembly (12944729)			
Push off test of detonator	50	3.2	4.5.5
Rotor and Gear Sector Assembly (Dwg. 12944730)			
Static push off test	50	3.2	4.5.13
Ogive (Dwg. 12944735)			
Salt Spray	8 1/	3.2	4.5.1
Centerplate (Dwg. 12944723)			
Deflection test	25	3.2	4.5.12
Bracket spring (Dwg. 8876226)			
Load test	25	3.2	4.5.2
Firing Pin Spring (Dwg. 12944749)			
Load test	25	3.2	4.5.2
Centerplate Assembly (Dwg. 12944734)			
Retention check of firing pin spring	125	3.2	4.5.16
Firing pin retraction	125	3.2	4.5.17
Dis-Lodging force test of hammerweight pins	25	3.2	4.5.15
Bracket pull off test from hammerweight pin	25	3.2	4.5.14
Escapement Assembly Less Setback Pin and Setback Spring (Dwg. 12944725 less 9287882 and 12944741)			
Arming time	125	3.2	4.5.18
Non-arming	125	3.2	4.5.19

NOTES:
1/ Sample taken from quantity of 25. Rejection shall be in accordance with MIL-A-8625, MIL-C-5541 (Finish 7.3.1).

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TABLE III. First Article Inspection - Continued

Fuze, PIBD, M549A1 Less Spitback and Components 2 of 2			
EXAMINATION OR TEST	NO. OF SAMPLE UNITS	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Escapement Assembly (Dwg. 12944725)			
Pull test of top plate	125	3.2	4.5.3
Fuze, PIBD, M549A1 Less Spitback (Dwg. 12944738)			
Transportation Vibration	400	3.3	4.5.21/ 4.520
Jolt	12	3.4	4.5.22/ 4.5.20
Jumble	96	3.5	4.5.23/4.5.20
Torque	125	3.2	4.5.27
Graze Sensitivity	150 Acc 5	3.6	4.5.24
	Rej 6		
All Arming	150 Acc 5	3.6	4.5.25
	Rej 6		
Non-Arming	100 Acc 0	3.7	4.5.26
	Rej 1		
Premature Burst	400 Acc 0	3.8	4.5.28
	Rej 1		
NOTES:			

4.4 Conformance inspection.

4.4.1 Inspection lot formation. Lot formation shall be in accordance with MIL-STD-1916. For the components identified below, each inspection lot of fuzes shall be produced with components complying with the following constraints:

a. Metal parts or components from one interfix number, from one manufacturer
b. Detonators from one interfix lot series and from one manufacturer.
c. Escapements from one interfix lot number, from one manufacturer

4.4.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for Critical, Major and Minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Inspection Level IV shall be used for all characteristics defined as Majors and Inspection Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.4.2.1 through 4.4.2.22. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER 12944746
4.4.2.1	Firing Pin			NEXT HIGHER ASSEMBLY 12944734
CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101	Profile of firing pin point	Level IV	3.2	Gage
102	Length from point to underside head (.245)	Level IV	3.2	Gage
103	Diameter of second largest diameter	Level IV	3.2	Gage
104	Height of firing pin head 1/	Level IV	3.2	Gage
105	Diameter of firing pin head 2/	Level IV	3.2	Gage
MINOR				
201	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33

NOTES:
1/ Refer to zone C-2 on Dwg. 12944746.
2/ Refer to zone B-3 on Dwg. 12944746.

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Setback Pin			9287882
				NEXT HIGHER ASSEMBLY 12944725
CATEGORY	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE / INSPECTION METHOD**
CRITICAL 1	Length of smallest diameter, min.	100%	3.2	Gage
MAJOR	None defined			
MINOR 201	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33
NOTES:				

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Setback Spring			12944741
				NEXT HIGHER ASSEMBLY 12944725
CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101	Push off test	80 1/	3.2	4.5.4
102	Diameter between tangs	Level IV	3.2	Gage
103	Tang broken, bent or missing	Level IV	3.2	Visual
104	Total length, min	Level IV	3.2	Gage
105	Diameter across legs, min	Level IV	3.2	Gage
106	Leg broken, bent or missing	Level IV	3.2	Visual
MINOR				
201	Total length, max.	Level II	3.2	Gage
202	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33

NOTES:
1/ Accept on zero (0) and reject on one (1).

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Bracket		12944724
			NEXT HIGHER ASSEMBLY 12944734

CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101	Inside width of bracket 1/	Level IV	3.2	Gage
102	Length from bottom of bracket to slot 2/	Level IV	3.2	Gage
103	Width at open end of bracket legs 1/	Level IV	3.2	Gage
104	Width of slot 3/	Level IV	3.2	Gage
105	Length from bottom of slot to tip of bracket legs 3/	Level IV	3.2	Gage
MINOR				
201	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33

NOTES:
1/ Refer to zone B-3/4 on Dwg. 12944724.
2/ Refer to zone D-6 on Dwg. 12944724.
3/ Refer to zone C-5 on Dwg. 12944724.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER 12944727
4.4.2.5	Verge Assembly			NEXT HIGHER ASSEMBLY 12944725
CATEGORY	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE / INSPECTION METHOD**
CRITICAL	None defined			
MAJOR				
101	Thickness of verge (see Dwg.12944744)	Level IV	3.2	Gage
102	Outside diameter of verge (see Dwg 12944744) 3/	Level IV	3.2	Gage
103	Length from top of largest verge arbor diameter to bottom of verge 2/	Level IV	3.2	Gage
104	Bearing diameters (2) (Dwg. 12944728) 1/	Level IV	3.2	Gage
105	Profile of verge where starwheel (8886370) is contacted (see Dwg. 12944725)	Level IV	3.2	Gage
106	Perpendicularity of verge with bearing diameter	Level IV	3.2	Gage
107	Torque test	Level III	3.2	4.5.8
108	Disassembly test	Level III	3.2	4.5.9
MINOR				
201	Lubricant missing	Level II	3.2	Visual
202	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33

NOTES:
1/ Refer to zone C/D-4, C/D-2 on Dwg. 12944728.
2/ Refer to zone D-3 on Dwg. 12944727.
3/ Refer to zone B-3 on Dwg. 12944744.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Pinion Assembly			8886368
				NEXT HIGHER ASSEMBLY 12944725
CATEGORY	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE / INSPECTION METHOD**
CRITICAL	None defined			
MAJOR				
101	Bearing diameter (2) (See Dwg. 8886369)	Level IV	3.2	Gage
102	Starwheel thickness (See Dwg. 8886370)	Level IV	3.2	Gage
103	True position of outside diameter of starwheel with bearing diameters	Level IV	3.2	Gage
104	Distance from top of pinion to top of bushing	Level IV	3.2	Gage
105	Perpendicularity of starwheel with bearing diameters	Level IV	3.2	Gage
106	Push off test	Level III	3.2	4.5.10
107	Torque test	Level III	3.2	4.5.11
MINOR				
201	Lubricant missing	Level II	3.2	Visual
202	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33
NOTES:				

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Rotor Assembly		12944729
			NEXT HIGHER ASSEMBLY 12944725

CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101	Detonator stake missing (2) min. required	Level IV	3.2	Visual
102	Detonator inverted	Level IV	3.2	Visual
103	Detonator below flush	Level IV	3.2	Gage/Probe
104	Discernible bulge or crack on detonator	Level IV	3.2	Visual
105	Push off test of detonator	Level III	3.2	4.5.5
106	Check test for detonator deterioration	As Required	3.2	4.5.7
MINOR				
201	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Rotor and Sector Gear Assembly			12944730
				NEXT HIGHER ASSEMBLY 12944725
CATEGORY	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE / INSPECTION METHOD**
CRITICAL	None Defined			
MAJOR				
101	Thickness of sector gear (Dwg. 9219872)	Level IV	3.2	Gage
102	Run out of sector gear	Level IV	3.2	Gage
103	Overall height	Level IV	3.2	Gage
104	Rotor post flush or below flush of sector gear	Level IV	3.2	Gage
105	Swaged rotor shaft flange flush or below flush of sector gear	Level IV	3.2	Gage
106	Static push off test of rotor	80 1/	3.2	4.5.13
MINOR				
201	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33

NOTES:
1/ Accept on zero (0) and reject on one (1).

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Ogive		12944735
			NEXT HIGHER ASSEMBLY 12944738

CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101	Salt spray	1/	3.2	4.5.1
102	Spherical radius	Level IV	3.2	Gage
103	Pitch diameter of thread	Level IV	3.2	Gage
104	Major diameter of thread, max	Level IV	3.2	Gage
105	Outside diameter, max	Level IV	3.2	Gage
106	Thread length	Level IV	3.2	Gage
107	True position of largest outside diameter with pitch diameter of thread	Level IV	3.2	Gage
108	Diameter of thread undercut	Level IV	3.2	Gage
109	Width of thread undercut	Level IV	3.2	Gage
110	Wall thickness 2/	Level IV	3.2	Gage
MINOR				
201	Color incorrect	Level II	3.2	Visual
202	Protective finish with bare spot exposing base metal except rack marks	Level II	3.2	Visual
203	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33

NOTES:
1/ Sampling and rejection shall be in accordance with MIL-A-8625, MIL-C-5541 (Finish 7.3.1).
2/ Refer to the corresponding note on Dwg. 12944735.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Centerplate			12944723
				NEXT HIGHER ASSEMBLY 12944734
CATEGORY	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE / INSPECTION METHOD**
CRITICAL	None Defined			
MAJOR				
101	Deflection Test	32 5/	3.2	4.5.12
102	Diameter of firing pin hole 1/	Level IV	3.2	Gage
103	Diameter of counter sink 2/	Level IV	3.2	Gage
104	Width of spoke 3/	Level IV	3.2	Gage
105	Corner radius or chamfer 4/	Level IV	3.2	Gage
MINOR				
201	Lubricant missing	Level II	3.2	Visual
202	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33

NOTES:
1/ Refer to zone B/C-3 on Dwg. 12944723.
2/ Refer to zone B-3 on Dwg. 12944723.
3/ Refer to zone B/C-7 on Dwg. 12944723.
4/ Refer to zone B-2/3 on Dwg. 12944723 and make sure each tab (3) is measured according to drawing.
5/ Accept on zero (0) and reject on one (1).

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.11	Bracket Spring			8876226
				NEXT HIGHER ASSEMBLY 12944734
CATEGORY	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE / INSPECTION METHOD**
CRITICAL	None Defined			
MAJOR				
101	Thickness	Level IV	3.2	Gage
102	Spring Load Test	Level III	3.2	4.5.2
MINOR				
201	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33
NOTES:				

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.12	Firing Pin Spring			12944749
				NEXT HIGHER ASSEMBLY 12944734
CATEGORY	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE / INSPECTION METHOD**
CRITICAL	None Defined			
MAJOR				
101	Thickness	Level IV	3.2	Gage
102	Spring Load Test	Level III	3.2	4.5.2
MINOR				
201	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33
NOTES:				

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.13	Centerplate Assembly		12944734
			NEXT HIGHER ASSEMBLY 12944738

CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101	Retention check of firing pin spring	Level III	3.2	4.5.16
102	Firing Pin Retraction Test	100%	3.2	4.5.17
103	Bracket pull off test from hammerweight pin	80 2/	3.2	4.5.14
104	Dis-Lodging force test of hammerweight pins	80 1/	3.2	4.5.15
MINOR				
201	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33

NOTES:
1/ Accept on zero (0) and reject on one (1). If the average separation force of the entire sample is less than the specified average on the applicable drawing the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916. If any one pin has a separation value less than the absolute minimum specified on the applicable drawing the lot shall also be rejected set aside for disposition in accordance with MIL-STD-1916.
2/ Accept on zero (0) and reject on one (1).

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Escapement Assembly Less Setback Pin and Setback Spring (1/)	SHEET 1 OF 1	12944725 (1/) NEXT HIGHER ASSEMBLY 12944738

CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL				
1	Arming time, minimum 2/	100%	3.2	AAIE/4.5.18
2	Non-arming 3/	100%	3.2	AAIE/4.5.19
3	Both spin locks/spring combination missing or out of position 4/	100%	3.2	AAIE/4.5.32
4	Setback pin hole blocked	100%	3.2	AAIE/4.5.32
MAJOR				
101	Arming time, max 2/	100%	3.2	AAIE/4.5.18
102	Pull off test of top plate	80 5/	3.2	4.5.3
103	Below flush of end face of posts after welding	Level IV	3.2	Gage
104	One spin lock/spring combination missing or out of position 4/	100%	3.2	AAIE/4.5.32
MINOR				
201	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33

NOTES:
1/ Excluding drawings 9287882 and 12944741.
2/ These tests shall be performed concurrently.
3/ This test shall be performed after successful completion of the arming time test.
4/ These tests may be performed concurrently.
5/ Accept on zero (0) and reject on one (1).

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.15	Escapement Assembly			12944725
				NEXT HIGHER ASSEMBLY 12944738
CATEGORY	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE / INSPECTION METHOD**
CRITICAL				
1	Setback pin protrustion	100%	3.2	AAIE/4.5.32
2	Setback spring missing	100%	3.2	AAIE/4.5.32
3	Setback pin missing or retracted	100%	3.2	AAIE/4.5.32
4	Rotor assembly in armed or partially armed position	100%	3.2	AAIE/4.5.32
5	Spin locks not engage in rotor	100%	3.2	AAIE/4.5.32
6	Setback pin not blocking rotor	100%	3.2	AAIE/4.5.32
7	Both spin locks/spring combination missing or out of position 1/	100%	3.2	AAIE/4.5.32
8	Verge/Pinion Assembly missing	100%	3.2	AAIE/4.5.32
MAJOR				
101	One spin lock/spring combination missing or out of position 1/	100%	3.2	AAIE/4.5.32
MINOR				
201	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33

NOTES:
1/ These tests may be performed concurrently.

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Escapement Assembly and Centerplate Assembly inserted into Ogive (1/)	SHEET 1 OF 1	12944725 (1/) NEXT HIGHER ASSEMBLY 12944738

CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL				
1	Setback pin missing	100%	3.2	AAIE/4.5.32
2	Setback spring missing	100%	3.2	AAIE/4.5.32
3	Rotor in Armed or partially armed position	100%	3.2	AAIE/4.5.32
MAJOR	None Defined			
MINOR				
201	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33

NOTES:
1/ Including drawings 12944734 and 12944735.

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PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.17	Ring			12944740
				NEXT HIGHER ASSEMBLY 12944747
CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101	Outside diameter	Level IV	3.2	Gage
102	Inside diameter	Level IV	3.2	Gage
103	Thickness	Level IV	3.2	Gage
MINOR				
201	Lubricant missing	Level II	3.2	Visual
202	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33
NOTES:				

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.18	Fuze, PIBD, Less Spitback		12944738
			NEXT HIGHER ASSEMBLY 12944747

CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL				
1	Rotor in armed or partially armed position	100%	3.2	AAIE/4.5.32
2	Transportation vibration	210 Acc-0 Rej-1	3.3	4.5.21/4.5.20
3	Jolt 1/	12 Acc-0 Rej-1	3.4	4.5.22/4.5.20
4	Jumble 1/	96 Acc-0 Rej-1	3.5	4.5.23/4.5.20
5	Non- Arming 2/	10 Acc-0 Rej-1	3.7	4.5.26
6	Premature (performed concurrently with M.102/103/C5)	210 Acc-0 Rej-1	3.8	4.5.28
MAJOR				
101	Torque test	80 3/	3.2	4.5.27
102	Graze sensitivity 2/ 4/	100 Acc-3 Rej-4	3.6	4.5.24
103	All Arming 2/	100 Acc-3 Rej-4	3.6	4.5.25
104	Pitch diameter of thread	Level IV	3.2	Gage
105	Major diameter of thread, max.	Level IV	3.2	Gage
106	True position of large cavity with small cavity	Level IV	3.2	Gage
107	Outside diameter, max.	Level IV	3.2	Gage
108	Total length, max.	Level IV	3.2	Gage
109	Perpendicularity of large outside diameter with crimped surface	Level IV	3.2	Gage
110	Crimp 360º	Level IV	3.2	Visual
MINOR				
201	Major diameter of thread, min.	Level II	3.2	Gage
202	Marking Missing, misleading or unidentifiable	Level II	3.2	Visual
203	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33

NOTES:
1/ If a detonator initiates during this test, the failure shall be considered a “Major” defect if the fuze is in the unarmed position.
2/ Units are subjected to transportation vibration test prior to this test.
3/ Accept on zero (0) and reject on one (1).
4/ Acceptance is based on requirements of paragraph 3.7 or disassembly of the round and examination of the M55 detonator for functioning in full armed position.

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PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.19	Test Vehicle, prior to Assembling Inert Body Assembly			12993605
				NEXT HIGHER ASSEMBLY
				12993604
CATEGORY	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE / INSPECTION METHOD**
CRITICAL	None Defined			
MAJOR 101	O-ring missing in groove	Level IV	3.2	Visual
MINOR 201	Evidence of poor workmanship	Level II	3.2	Visual/4.5.33
NOTES:				

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.20	Body Loading Assembly, Inert (Test Vehicle)		12993606
			NEXT HIGHER ASSEMBLY 12993605

CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101	Depth to liner	Level IV	3.2	Gage
102	Diameter of rotating bands	Level IV	3.2	Gage
103	True position of liner	Level IV	3.2	Gage
104	Cap missing, loose or torn exposing inert material	Level IV	3.2	Visual/Manual
105	Specific gravity, min.	Level III	3.2	4.5.31
MINOR				
201	Thread damaged	Level II	3.2	Visual
202	Evidence of poor workmanship	Level II	3.2	Visual/4.5.33

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.21	Capsule Assembly (Test Vehicle)			12993607
				NEXT HIGHER ASSEMBLY 12993604
CATEGORY	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE / INSPECTION METHOD**
CRITICAL	None Defined			
MAJOR 101	Weight of powder	100%	3.2	Gage
MINOR 201	Evidence of poor workmanship	Level II	3.2	Visual/4.5.33
NOTES:				

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.22	Test Vehicle 1/	SHEET 1 OF 1	12993604 12993605 NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101	Chamber gage failure	100%	3.2	Gage/9202253, 9202254, 9202255
102	Pull test of projectile	Level III	3.2	4.5.29
103	Air pressure test	Level III	3.2	4.5.30
104	O-ring exposed	Level IV	3.2	Visual
105	Diameter of rotating bands	Level IV	3.2	Gage
106	True position of capsule seat to minor diameter	Level IV		Gage
MINOR				
201	Total length	Level II	3.2	Gage
202	Marking Missing, misleading or unidentifiable	Level II	3.2	Visual
203	Evidence of poor workmanship	Level II	3.9	Visual/4.5.33

NOTES:
1/ All test vehicle components shall meet the Conformance Inspections slated in MIL-DTL-50863 and MIL-DTL-60982.

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4.5 Methods of inspection

4.5.1 Salt spray. The specified part shall be subjected to the salt spray test in accordance with ASTM B117. This test shall be conducted for 96 hours using a salt solution of 5 percent. This test is a destructive test. Parts so tested shall not be returned to the lot. Sampling and rejection shall be in accordance with MIL-A-8625

4.5.2 Load requirement at each position. The applicable part shall be placed in an approved fixture and the load requirement at each position shall be measured. Observation shall be made for conformance with the applicable drawing requirement. This is a non destructive test. Parts so tested may be returned to the lot.

4.5.3 Pull off test of top plate. An empty escapement assembly (escapement housing, Dwg. 12944726, and staked top plate, Dwg. 12944732 only) shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied. Observation shall be made for conformance with the applicable drawing requirement. This is a destructive test. Parts tested shall not be returned to the lot.

4.5.4 Push off test of setback spring. The setback spring shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied to either a setback pin, Dwg. 9287882, or an equivalent test item. Observation shall be made for conformance with the applicable drawing requirement. This is a destructive test. Setback pins and setback springs subjected to this test shall not be returned to the lot.

4.5.5 Push off test of detonator assembly. The rotor assembly shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied to the detonator, as a minimum. The push off force shall not exceed 10% of the specified minimum value. The push off force shall be applied uniformly over the surface of the detonator. This test is considered a non destructive test within the specified limits. Parts so tested may be returned to the lot.

4.5.6 Ammunition lot numbering. Visually verify that an ammunition lot number has been applied to each lot as described in MIL-STD-1168.

4.5.7 Check test for deterioration of detonators. If the total time between original acceptance of any lot and the assembly of that lot into the escapement assembly exceeds two years, or if the detonators have been subjected to adverse conditions, however brief, at any time since the previous tests, the detonator lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in MIL-D-14978 immediately before the detonator lot is assembled into the escapement assembly. This test shall be performed by the contractor on detonators selected by the Government inspector at the facility assembling the detonators into the escapement assembly.

4.5.8 Torque test of verge assembly. The verge assembly shall be placed in an approved fixture and the torque specified on the applicable drawing shall be applied, as a minimum. The torque applied to the verge assembly shall not exceed 20% of the specified minimum value. The bearing surfaces of the verge assembly shall not be utilized for retention of the assembly during the torque test if the parts are intended to be returned to the lot. This test is considered a non destructive test within the specified limits. Parts so tested may be returned to the lot.

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4.5.9 Disassembly test of verge assembly. The verge assembly shall be placed in an approved fixture and the disassembly force specified on the applicable drawing shall be applied, as a minimum. The disassembly force shall not exceed 10% of the specified minimum value. The disassembly force shall be applied uniformly over the surface of the verge. This test is considered a non destructive test within the specified limits. Parts so tested may be returned to the lot.

4.5.10 Push off test of pinion assembly. The pinion assembly shall be placed in an approved fixture and the push off force specified on the applicable drawing shall be applied, as a minimum. The push off force shall not exceed 10% of the specified minimum value. The push off force shall be applied uniformly over the surface of the star wheel. This test is a considered non destructive test within the specified limits. Parts so tested may be returned to the lot.

4.5.11 Torque test of pinion assembly. The pinion assembly shall be placed in an approved fixture and the torque specified on the applicable drawing shall be applied, as a minimum. The torque applied to the pinion assembly shall not exceed 20% of the specified minimum value. The bearing surfaces of the pinion assembly shall not be utilized for retention of the assembly during the torque test if the parts are intended to be returned to the lot. This test is considered a non destructive test within the specified limits. Parts so tested may be returned to the lot.

4.5.12 Deflection test of centerplate. The centerplate shall be placed in an approved fixture and the minimum force specified on the applicable drawing shall be applied to the firing pin hole. The total deflection shall be recorded. This is a destructive test. Parts so tested shall not be returned to the lot.

4.5.13 Static push test of rotor and gear sector assembly. The rotor and gear sector assembly shall be placed in an approved fixture and the push off force specified on the applicable drawing shall be applied, as a minimum. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.14 Bracket pull off test from hammerweight pin. The centerplate assembly shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied to the bracket. During this test the bracket shall not be separated from the hammerweight pin or have any evidence of separation. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.15 Dis-Lodging force test of hammerweight pins. The centerplate assembly shall be placed in an approved fixture and the force necessary to push out the hammerweight pin at each of the three legs shall be recorded. The force application orientation shall be in accordance with the applicable drawing. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.16 Retention check of firing pin spring on centerplate assembly. After firing pin spring (Dwg. 12944749) is assembled to firing pin (Dwg. 12044746) as shown on centerplate assembly (Dwg. 12944734), the centeplate assembly shall be placed in an approved fixture and only the three long firing pin spring legs shall support the weight of the centerplate assembly. During this test, the bottom of the centerplate (Dwg. 12944723) shall be suspended from surface of the fixture. This test is a non-destructive test. Parts so tested may be returned to the lot.

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4.5.17 Firing pin retraction of centerplate assembly. After completion of the retention check of firing pin spring on centerplate assembly of 4.5.16 the firing pin retraction shall be checked. The centerplate assembly shall be placed in an approved fixture and the position of the firing pin point in spin specified on the applicable drawing shall be checked. During spinning, no external force shall be applied to the firing pin to check retraction. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.5.18 Escapement arming time. The escapement assembly, less setback pin and spring, shall be placed in an approved fixture and rotated at the speed specified on applicable drawing. Observation shall be made for conformance with the applicable drawing requirement. This is a non-destructive test. Assemblies subjected to this test may be returned to the lot.

4.5.19 Escapement non-arming. After successful completion of the test specified in 4.5.18, the escapement assembly, less setback pin and setback spring, shall be reset and placed in an approved fixture and rotated at the non-arming speed specified on the applicable drawing. Observation shall be made for a blocked setback pin hole. This is a non-destructive test. Assemblies subjected to this test may be returned to the lot.

4.5.20 X-ray examination. Examination shall be performed with X-ray equipment or any other method acceptable to the contracting officer. Radioscopic examination equipment, procedures, and personnel shall be in accordance with the requirements of ASTM E1255. This inspection shall be conducted in two (2) planes: from the top of the fuze assembly down; and from the side. The side view shall show the setback pin adjacent to the rotor. After successful completion of this inspection, the fuze assembly shall be subjected to the transportation vibration, jolt or jumble test.

4.5.21 Transportation vibration. Prior to the transportation vibration the fuze assemblies shall be x-rayed in accordance with 4.5.20. The fuze assemblies packed in their regular shipping containers shall be tested in accordance with MIL-STD-331 (Test No.B2, with vibration levels conforming to 6.1.2.1 of Test B1). The test shall be performed at ambient temperature only. After completion of the test, the fuze assemblies shall be x-rayed as specified in 4.5.20. Observation shall be made for conformance with the requirements specified in 3.3.

4.5.22 Jolt. Prior to the jolt test the fuze assemblies shall be x-rayed in accordance with 4.5.20. The fuze assembly shall be tested in accordance with Test No. A1-1 of MIL-STD-331. After completion of the test, the fuze assembly shall be x-rayed as specified in 4.5.20. This is a destructive test. Parts so tested shall not be returned to the lot. Observation shall be made for conformance with the requirements specified in 3.4.

4.5.23 Jumble. Prior to the jumble test the fuze assemblies shall be x-rayed in accordance with 4.5.20. The fuze assembly shall be tested in accordance with test No. A2-1 of MIL-STD-331. After completion of the test, the fuze assembly shall be x-rayed as specified in 4.5.20. This is a destructive test. Parts so tested shall not be returned to the lot. Observation shall be made for conformance with the requirements specified in 3.5.

4.5.24 Graze sensitivity. After successful completion of the transportation vibration test, a spitback assembly (Dwg. 9287862) shall be assembled to the fuze, PIBD M549A1, less spitback (Dwg. 12944738) in accordance with Dwg. 12944747. In turn, this assembly shall be assembled to the test vehicle w/crtg case M169 for M549A1 fuze (Dwg. 12993605) in accordance with Dwg. 12993604. The assembled cartridges shall be fired from a MK19, Mod 3, machine gun at an

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elevation that will cause it to impact at approximately 200 ± 5 meters from the weapon. The impact medium shall be 10 inches, minimum, of coarse sand which shall be graded at an angle of approximately 25° with the horizontal. The sand shall be raked or the weapon transverse between shots so that the impacts are not overlapping. If a round falls outside the specified area, it shall be declared a "no test". When results from the functioning test are reported, the number of "no tests" shall also be reported. The velocity of the sample cartridges shall be tested and measured in accordance with MIL-DTL-50863 (790 ± 10 feet per second (fps) with a standard deviation not exceeding 12.0 fps or less). Observation shall be made for conformance with the requirements specified in 3.6 and the velocity requirements of MIL-DTL-50863.

4.5.25 Fuze all arming test. After successful completion of the transportation vibration test, a spitback assembly (Dwg. 9287862) shall be assembled to the fuze, PIBD M549A1, less spitback (Dwg. 12944738) in accordance with Dwg. 12944747. In turn, this assembly shall be assembled to the test vehicle w/crtg case M169 for M549A1 fuze (Dwg. 12993605) in accordance with Dwg. 12993604. The assembled cartridge shall be fired from a MK19 Mod 3 machine gun to impact at approximately 195 ± 5 feet against a vertical 1/2 or 5/8 inch piece of insulation board per ASTM C 208 or approved equals. If a round fails to hit the insulation board, it shall be declared a "no test". When the results from the functioning test are reported, the number of "no tests" shall also be reported. The velocity of the sample cartridges shall be tested and measured in accordance with MIL-DTL-50863 (790 ± 10 feet per second (fps) with a standard deviation not exceeding 12.0 fps or less). Observation shall be made for conformance with the requirements specified in 3.6 and the velocity requirements of MIL-DTL-50863.

4.5.26 Fuze non-Arming test. After successful completion of the transportation vibration test, a spitback assembly (Dwg. 9287862) shall be assembled to the fuze, PIBD M549A1, less spitback (Dwg. 12944738) in accordance with Dwg. 12944747. In turn, this assembly shall be assembled to the test vehicle w/ctrg. case M169 for M549A1 fuze (Dwg. 12993605) in accordance with Dwg. 12993604. The assembled cartridge shall be fired from a MK19, Mod 3, machine gun to impact at 60 feet against a vertical 1/2 or 5/8 inch thick piece of insulation board. If a round fails to hit the target or hits the target and does not function and does not function on impact on a vertical 1/2" thick mild steel plate placed at 200 feet from the launcher, it shall be declared a "no test". When results from the non-functioning test are reported, the number of "no tests" shall also be reported. The velocity of the sample cartridges shall be tested and measured in accordance with MIL-DTL-50863 (790 ± 10 feet per second (fps) with a standard deviation not exceeding 12.0 fps or less). Observation shall be made for conformance with the requirement specified in 3.7 and the velocity requirements of MIL-DTL-50863.

4.5.27 Torque test. Sample shall be placed in an approved fixture and the torque specified on the applicable drawing shall be applied. Observation shall be made for conformance with the applicable drawing requirement. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.28 Premature burst. This test shall be conducted concurrently with the tests specified in 4.5.24, 4.5.25, and 4.5.26. Observation shall be made for conformance with the requirements specified in 3.8.

4.5.29 Pull test of projectile. The cartridge shall be placed in an approved fixture and the axial tension specified on the applicable drawing shall be applied. Cartridge case assemblies may be reused after undergoing a Government approved rework provided the same crimping method is used. Projectile assemblies if undamaged may be reused. The cartridge shall be pulled until separation occurs and the data recorded.

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4.5.30 Air pressure. The cartridge assembly shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the required air pressure. Observation shall be made for conformance with the applicable drawing requirement.

4.5.31 Specific gravity of inert material. The specific gravity of the inert material shall be determined as follows:

Weigh the empty body assembly in grams and record as We. Fill the empty assembly with water at 75 degrees + 2 degrees F., weigh in grams and record the weight as Ww. Determine the volume by subtracting We from Ww. Load the assembly with inert material, weigh in grams and record the weight as W1. Determine the inert material weight by subtracting We from W1. Calculate the specific gravity as follows:

$$\text{Specific gravity} = \frac{W1 - We}{Ww - We}$$

Observation shall be made for conformance with the applicable drawing requirement.

4.5.32 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the Inspection Method Column in the tables starting with paragraph 4.4.2.1. Inspection equipment used for acceptance of product shall be capable of repeatable measurements, by various experienced inspection/test personnel, to an accuracy of 10% or better of the total tolerance of the characteristic being inspected. Any automated acceptance inspection equipment (AAIE) prove-out shall be performed in accordance with MIL-A-70625 (see 6.7).

4.5.33 Workmanship. Visually verify that all parts and assemblies meet requirements of paragraph 3.9.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fuze assemblies covered by this specification are intended for use with the Cartridge, 40MM: HEDP, M430A1.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

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b. The title, number (including revision level) and date of all documents referenced (see Section 2).
c. Requirements for submission of first article sample.
d. Requirement for submission of design verification sample, if applicable.
e. Requirement for submission of inspection equipment designs.
f. Applicable national stock number.
g. Packaging requirements, if other than specified in Section 5.
h. Serialization requirements, if applicable.
i. Certificate of conformance for each lot or shipment of product, if applicable.
j. Government-Furnished Material or equipment, if applicable.
k. Provisions for critical characteristics controls.
l. Critical Inspection Equipment Requirement (See 6.7)

6.3 Submission of inspection equipment designs for approval. Submit equipment designs as required to Commander, ARDEC, ATTN: AMSRD-AAR-QEM-D, Picatinny, NJ 07806-5000. Request letter of submittal should state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph. Inspection equipment, inspection method and the frequency of inspection should be submitted for approval to ARDEC, ATTN: AMSRD-AAR-QEM-D, Picatinny, NJ 07806-5000.

6.4 Ammunition lot numbers. Ammunition requires ammunition data cards in accordance with MIL-STD-1168.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 Definitions.

6.6.1 Metal defective. Nick, crack, dent or sharp edge which might interfere with the proper assembly or end item performance.

6.6.2 Critical defect. A critical defect is a defect that judgment and experience indicate is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product; or a defect that judgment and experience indicate is likely to prevent performance of the tactical function of a major end item.

6.6.3 Major defect. A major defect is a defect, other than critical, that is likely to result in failure or to reduce materially the usability of the unit of product for its intended purpose.

6.6.4 Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose or is a departure from established standards having little bearing on the effective use or operation of the unit.

6.6.5 Subject term (key words) listings.

Cartridge, 40mm, HEDP, M430A1
Detonator, Stab, M55
Gun, Machine, 40MM, MK19, Mod 3

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6.7 Critical Inspection Equipment

6.7.1 System Reliability. The contractor should provide for each defect identified as critical, a manufacturing and inspection system that assures no more than one in a million contain the defect. This should be assured by controlling the maximum defect rate produced and the error rate of the inspection equipment such that the product of the two terms is less than one in a million.

6.7.2 Maximum defect rate produced. The maximum defect rate produced should be defined as the largest defect rate expected for the characteristic of concern on a monthly or lot basis. The contractor should establish the maximum defect rate, however, it may not exceed one percent without approval of the procuring contracting officer. Once established, the contractor should monitor the defect rate to assure it does not exceed the maximum rate allotted. If the established maximum defect rate is exceeded, the production of that feature should be terminated until the cause is determined and corrected. Additionally, all products for that period of time should be rejected and re-inspected.

6.7.3 Maximum error rate of the inspection system. The error rate of the inspection equipment should be defined as the expected ratio of the number of defective parts accepted to the number of defective parts inspected by the equipment. The contractor, to meet the system reliability as defined in the contract. However, it may not exceed 1/500 without approval of the procuring contracting officer. Based on the maximum error rate defined for the equipment, the contractor should develop a demonstration test procedure to demonstrate the error rate of the equipment. The test should be performed using defective parts or reject standards. No part or standard should be accepted during the test. If a part or standard is accepted, the cause for failure should be isolated and corrected and the test rerun. Unless otherwise specified in the detailed item specification, the minimum number of test samples to be run should be equal to 0.7 divided by the error rate (e.g. If the required error rate is 1/1000, the sample size would be 0.7 divided by (1/1000) = 700.)

6.7.4 Periodic verification. Periodic verification of the system error rate should be performed.

6.7.5 Redundant inspection equipment. The contractor may elect to use redundant inspection equipment in lieu of using a single piece of equipment with a very low error rate. Where redundant equipment is used, the inspection system error rate is defined as the error rates of the multiple pieces of equipment multiplied together.

6.7.6 Automated acceptance inspection equipment. Equipment for which no, or minimal, human involvement is required in the acceptance determination. This would include equipment employing probes/sensors/transducers that are automatically manipulated to perform measuring and detection functions. If readout is provided, the equipment will be categorized as automated even if operator interpretation is needed in the accept/reject decision.

6.8 Revisions/Amendments History. The following is a record of changes incorporated into this document:

ECP	Description	Date Approved
R04Q2017	Initial Release of M549A1 Detail Specification	Aug 2004

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6.9 Interim Specification This interim specification has been prepared at Program Manager request to document in ASSIST Online the version of a draft specification that was placed on contract in 2004. This interim specification has been dated to match the date of the draft specification that was placed on contract. It has been modified slightly to comply with MIL-STD-961 requirements for format, but not content. A fully coordinated revision, compliant with both MIL-STD-961 format and content requirements, will be approved for publication.

6.10 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issues due to the extent of the changes.

Preparing activity:

Army-AR

(Project 1390-0893-000)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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